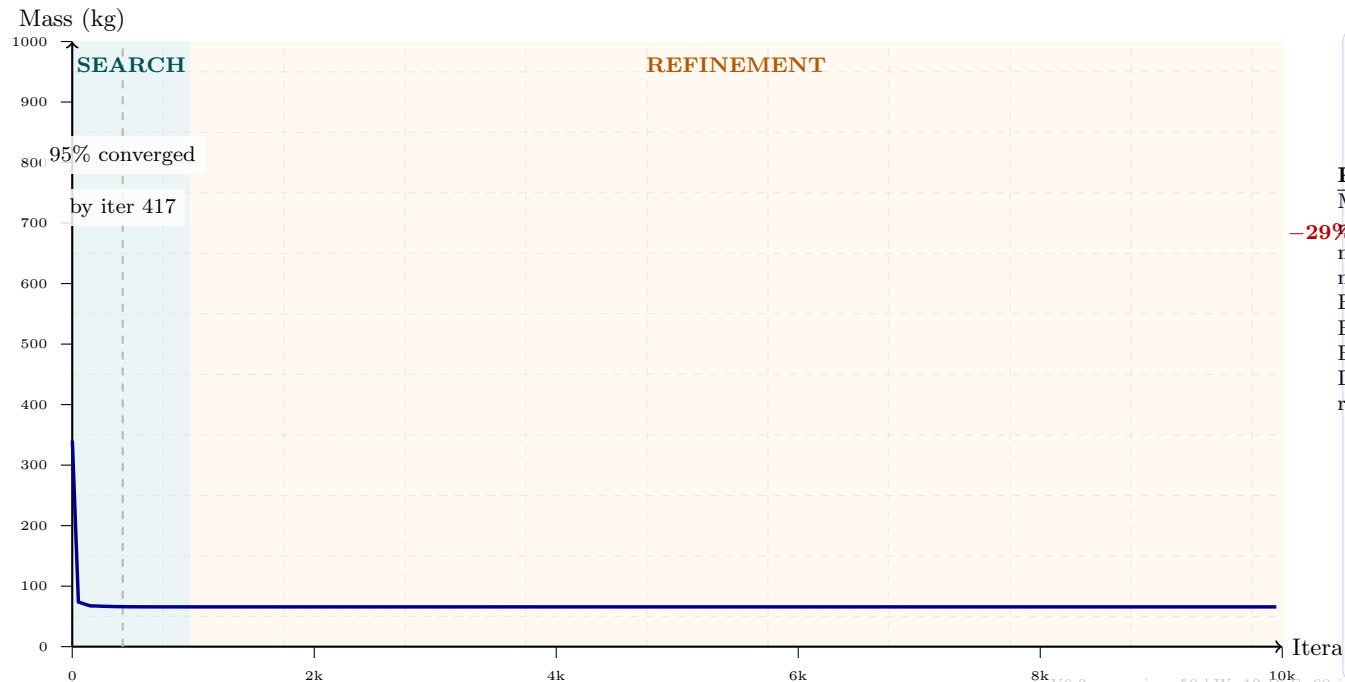




Parameter trends

V6.2 → V6.3 (12th DoF)

Parameter	V6.2	V6.3
Mass	74.2 kg	52.6 kg
n_lines	12	7
n_expansion	1	6
n_rings	9	8
Blade scale λ	1.0 (fixed)	0.200
Blade tip r	10.6 m	1.8 m
Bank angle	45°	35° (cap)
D _o top	95 mm	83 mm
r_hub	5.4 m	5.4 m

The blade scale λ

Freed from fixed $\lambda=1.0$, the optimizer

chose $\lambda=0.200$:

- Blades 5× shorter
- Blades 125× lighter (λ^3)
- 6 small rotors instead of 1 big

Bounds at limits

$\lambda=0.200$ (min)

n_exp=6 (max)

bank=35° (safety cap)

t/D=0.01 (min)

→ True optimum likely <50 kg.

Widen bounds in V6.4.